A fixed-charge covariance tangent

An exact two-deficit channel expansion

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Abstract

The normalized covariance of the overlapping complex p-body model is an exact path of charge-changing completely positive maps. We expand this path to first order around a scalar reference, with an explicit second-order remainder valid even when a reference factor vanishes. Previously established pure-Wick deficit bounds then give a same-copy local-size tangent with error of order p^{-1} at each fixed degree and finite Schatten order. This isolates one input to the central-sector neutral-memory problem; the full neutral transfer is not asserted here.

1 Model and inherited inputs

Use standard CAR, $\{c_i, c_j^\dagger\} = \delta_{ij}$, on $\mathcal{H}_m = \wedge^m \mathbb{C}^N$, and the same static Gaussian couplings on all sectors. Set $Z_v = \mathcal{W}_v(H^v)$, where $\mathcal{W}_v$ is projection onto top coupling chaos, and

$$\|X\|_{r,m} = (\mathbb{E}\,\mathrm{tr}_m\,|X_m|^r)^{1/r}, \qquad \mathrm{tr}_m = \binom{N}{m}^{-1} \mathrm{Tr}_{\mathcal{H}_m}.$$

The covariance and its scalar normalization are

$$\Phi_m(X) = w_N \sum_{|I|=|J|=p} C_I^\dagger C_J X C_J^\dagger C_I, \qquad C_m = w_N \binom{m}{p}\binom{N-m+p}{p}.$$

The energy normalization is fixed, $\hbar = 1$; no time rescaling is used in this note. For neutral block families define, before restricting,

$$TX = \sum_i c_i^\dagger X c_i, \quad SX = \sum_i c_i X c_i^\dagger, \quad D = Q - T, \quad \bar{D} = N - Q - S, \quad L = (D + \bar{D})/2.$$

We inherit the exact covariance path and the pure-Wick first and second deficit bounds from [4, 3, 5]. The literature context is fixed-charge complex SYK [1]. The SU(2) size construction of [2] distinguishes a representation rank from local Clifford size; its Brownian dynamics are not used here.

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2 Exact expansion, including zero reference factors

Write $t_k = T/k$ and $s_k = S/(N-k)$ for the maps with target charge k. They are unital and preserve normalized traces between adjacent sectors, hence contract every normalized Schatten norm, also after disorder averaging. Expansion of their Kraus products gives [4]

$$\Psi_m := \Phi_m / C_m = t_m \cdots t_{m-p+1} s_{m-p} \cdots s_{m-1}.$$

Let $\ell = 2p$ and $K_1 \cdots K_\ell$ denote this outer-to-inner path. The target charges are $k_0 = m, k_1 = m-1, \ldots, k_p = m-p, k_{p+1} = m-p+1, \ldots, k_\ell = m$. Its denominators are

$$(d_1, \ldots, d_\ell) = (m, m-1, \ldots, m-p+1;\ N-m+p, \ldots, N-m+1).$$

For a constant b put $a_j = 1 - b/d_j$, $\theta = \prod_j a_j$, and $\Delta_j = b - D$ on a t step, $\Delta_j = b - \bar{D}$ on an s step. The exact identity at each step's target is $K_j X = a_j X + d_j^{-1} \Delta_j X$.

Proposition 1. *For every neutral block family,*

$$\Psi_m X_m = \theta X_m + \sum_{j=1}^{\ell} \frac{\prod_{h \neq j} a_h}{d_j} (\Delta_j X)_m + \mathcal{R}_m(X), \tag{1}$$

$$\mathcal{R}_m(X) = \sum_{i<j} \frac{\prod_{\substack{h>i \\ h \neq j}} a_h}{d_i d_j} \left(K_1 \cdots K_{i-1} \Delta_i \Delta_j X\right)_m. \tag{2}$$

In the remainder the composed deficits act on the full family before the prefix path is applied. No division by a_j occurs. If $0 \le b \le d_{\min} := \min_j d_j$, then for $1 \le r < \infty$,

$$\|\mathcal{R}_m(X)\|_{r,m} \le \binom{\ell}{2} d_{\min}^{-2} \sup_{\substack{k \text{ on path} \\ \sigma,\tau \in \{D,\bar{D}\}}} \|(b-\sigma)(b-\tau)X\|_{r,k}\,. \tag{3}$$

Proof. First telescope against the scalar path:

$$\Psi_m X_m - \theta X_m = \sum_j \frac{\prod_{h>j} a_h}{d_j} (K_1 \cdots K_{j-1} \Delta_j X)_m.$$

Apply the same telescope to each prefix acting on the full family $\Delta_j X$. Its scalar part gives the first sum in (1); its difference gives (2), with the displayed order $\Delta_i \Delta_j$. Finally $0 \le a_h \le 1$ and prefix contraction bound every term by its source-sector norm. □

3 Fixed-degree local-size tangent

Take $\lambda_N = p^2/N$ in a compact subset of $(0,\infty)$, $p \to \infty$, $|m - N/2| \le L_0 p$, and fixed v, r, L_0. Set $b = vp$ and $\theta_{v,m} = \prod_j (1 - vp/d_j)$. Eventually $d_{\min} > vp$. Define

$$\zeta_T = p \sum_{j=0}^{p-1} (m - j - vp)^{-1},$$
$$\zeta_S = p \sum_{j=0}^{p-1} (N - m + p - j - vp)^{-1}.$$

Then the actual same-disorder residual satisfies the exact formula

$$\begin{aligned} A_v &:= p(\Phi_m - C_m \theta_{v,m}) Z_v \\ &= C_m \theta_{v,m} \{\zeta_T (vp - D) Z_v + \zeta_S (vp - \bar{D}) Z_v\}_m + p C_m \mathcal{R}_m(Z_v). \end{aligned} \tag{4}$$

For $D_* = \max_j |N/2 - (d_j - vp)| < N/2$, subtraction of the reciprocal $2/N$ term by term gives

$$|\zeta_\sigma - 2\lambda_N| \le \frac{p^2 D_*}{(N/2)(N/2 - D_*)}, \qquad \sigma = T, S. \tag{5}$$

Since $D_* = O_{v,L_0}(p)$ and $N \asymp p^2$, this is $O(p^{-1})$. The remainder coefficient in (3), after multiplying by p, is

$$\frac{p\binom{2p}{2}}{d_{\min}^2} = O(p^3/N^2) = O(p^{-1}). \tag{6}$$

Corollary 1. *With the pure repeated-deficit bounds of [3, 4],*

$$\boxed{\|A_v - 4\lambda_N C_m \theta_{v,m} (vp - L) Z_v\|_{r,m} = O_{v,r,L_0}(p^{-1}).} \tag{7}$$

The constant is uniform on the stated central charge window.

Proof. For fixed v the inherited bounds control both first deficits and all ordered second deficits at every charge visited by the path. These charges lie in an enlarged compact central window. C_m is uniformly bounded in the fixed energy normalization, and $0 \le \theta_{v,m} \le 1$. Subtract the terms with both $\zeta_\sigma = 2\lambda_N$ in (4) and use (3), (5), (6). Their sum is precisely $4\lambda_N (vp - L) Z_v$. □

This is a strong estimate on the same microscopic coupling sample, with annealed norms. The finite parameters $\lambda_N, C_m, \theta_{v,m}$ are retained: their convergence to limiting constants need not have rate p^{-1}. The constant in (7) is not claimed uniform in v.

4 Evidence and the next interface

The saved exact checker uses Python 3.14.5 and rational `fractions.Fraction` arithmetic. It tests all 96 neutral matrix units for $N = 2, 3, 4$, path lengths with $p \le 2$, admissible target charges, and $b = 0, 1/2, d_{\min}$. It records 1,824 path identities and 1,824 normalized S_2 remainder inequalities, including 608 zero-reference cases and 904 nonzero remainders. Sixty scalar rows use $N = 4p^2$, $p = 4, 8, 16, 32, 64$, central offsets $-2, 0, 2$ and $v = 1, 2, 3, 5$.

These checks test the finite algebra and scalar bounds, not the Gaussian asymptotic theorem. The first attempt accidentally made an empty product an integer and introduced floating-point division. Its failure and code are preserved; explicit rational seeds correct that implementation. The successful science output is not rerun or overwritten.

The exact neutral coefficient still has two pieces,

$$T_j = A_j + (C_m\theta_{j,m} - \Phi_m)\eta_j, \qquad \eta_j = p[Z_j - Q_{j,m}(H)].$$

Here $Q_{j,m}$ is the existing covariance-normalized Jacobi polynomial. Equation (7) controls the first piece. It does not identify the action of Φ_m on the lower-chaos composite descendants in the second. The next task is to audit the earlier compact-central rank-series bridge [6] and the rank-to-size identity [7], then combine fixed-degree transfer with sectorwise tails and exact energy pinching. The single-selected functional must retain its local-size boundary terms; the D-only density cannot be substituted for general neutral words. Actual memory return and the full selected bilocal remain subsequent interfaces. No bulk spatial/mass or finite-G dictionary is derived in this note.

References

[1] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983.

[2] M. Tarakemeh and S. Xu, *Non-Abelian operator size distribution in charge-conserving many-body systems*, arXiv:2511.16634v1.

[3] Project derivation, `notes/susskind-bilocal-fourth-tail-diagram-draft.md`, §§1–4.

[4] Project derivation, `notes/susskind-bilocal-fourth-tail-operator-draft.md`, §§2–4.

[5] Project draft, `manuscript/susskind-sector-gradient-tail/main.tex`.

[6] Project draft, `manuscript/susskind-memory-neutral-marked-series/main.tex`.

[7] Project draft, `manuscript/susskind-memory-size-bridge/main.tex`, §1.